NORTHWESTERN UNIVERSITY

Don Mariano Marcos Avenue, Laoag City, 2900, Ilocos Norte, Philippines

PROJECT PROPOSAL	OSA-F05A

I. Project Proposal Title:

Robotics Seminar

II. Sponsor / Organization:

Icons

III. Date & Venue:

Sep 12, 2025 | SC

IV. Target Participants:

Computer Science Students

V. Background / Rationale:

Robotics has rapidly evolved from a niche field into a transformative force across industries such as manufacturing, healthcare, logistics, and everyday consumer applications. Advances in artificial intelligence, machine learning, and sensor technologies have significantly enhanced the capabilities of modern robotic systems, enabling them to perform complex tasks with greater precision, efficiency, and autonomy.

As organizations increasingly adopt automation to remain competitive, there is a growing need for individuals to understand the fundamentals, applications, and implications of robotics. However, many students and professionals have limited exposure to current developments and practical knowledge in this field.

This robotics seminar is organized to bridge that gap by providing participants with foundational insights, real-world applications, and emerging trends in robotics. It aims to foster interest, encourage innovation, and equip attendees with the knowledge necessary to engage with and contribute to the rapidly advancing technological landscape.

VI. Objectives:

1. To introduce the fundamental concepts of robotics, including key components such as sensors, actuators, and control systems.
2. To familiarize participants with common robotics tools, platforms, and programming basics.
3. To demonstrate real-world applications of robotics across various industries.
4. To develop participants' problem-solving and critical thinking skills through hands-on activities.
5. To provide practical experience in assembling and/or programming a simple robotic system.
6. To encourage collaboration and teamwork through group-based challenges.
7. To inspire interest in robotics and emerging technologies for future learning or career exploration.

VII. UNSDGs:

Quality Education

VIII. Approach / Process:

1. Pre-Seminar Preparation

Identify target participants and their skill level (e.g., beginners, students, professionals)
Prepare training materials, presentation slides, and robotics kits (if applicable)
Set up the venue, including technical requirements such as laptops, projectors, and internet access

2. Opening Session (9:00 AM – 9:30 AM)

Registration and welcome remarks
Overview of seminar objectives, agenda, and expected outcomes
Brief introduction to robotics and its relevance today

3. Fundamentals Session (9:30 AM – 11:00 AM)

Introduction to basic robotics concepts (sensors, actuators, control systems)
Overview of common platforms and tools
Q&A to clarify foundational understanding

4. Demonstration & Case Studies (11:00 AM – 12:30 PM)

Live demonstration of simple robotic systems
Real-world applications (e.g., industrial automation, healthcare robots)
Discussion of current trends and innovations

Lunch Break (12:30 PM – 1:30 PM)

5. Hands-On Workshop (1:30 PM – 3:30 PM)

Participants work in small groups with robotics kits or simulations
Guided activity (e.g., assembling a basic robot or programming simple movements)
Facilitators provide step-by-step support

6. Group Activity / Mini Challenge (3:30 PM – 4:30 PM)

Teams complete a simple robotics challenge (e.g., obstacle navigation, line-following task)
Encourages collaboration, problem-solving, and creativity

7. Presentation & Feedback (4:30 PM – 5:00 PM)

Groups present their outputs or demonstrate their robots
Facilitators provide feedback and key takeaways

8. Closing Session (5:00 PM – 5:30 PM)

Summary of key learnings
Open forum for questions
Distribution of certificates and evaluation forms

IX. Expected Outcomes:

Demonstrate a basic understanding of robotics concepts, including key components and how they function together.
Identify common applications of robotics in real-world settings such as industry, healthcare, and automation.
Apply fundamental programming or assembly skills to operate a simple robotic system.
Successfully complete a guided hands-on activity or mini challenge involving a robotic task.
Exhibit improved problem-solving, critical thinking, and teamwork skills through group activities.
Show increased interest and confidence in exploring robotics and related technologies.
Provide feedback and reflections on their learning experience for future improvement of similar programs.

X. Budget

a. Proposed Budget: ₱ 4,400.00

#	Item	Qty	Unit Price	Amount
1	snacks	100	₱ 44.00	₱ 4,400.00
			TOTAL	**₱ 4,400.00**

c. Source of Funding: Icons Fund

APPROVAL SIGNATURES

Renier Candido
Org President

Rowell Casil
Dept Head

Donald Buted
Org Adviser

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College Dean